MIL-C-24643/50B
3 October 1986
SUPERSEDING
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8 November 1985
MIL-C-24643/50
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, FOUR CONDUCTOR, TYPE LSFNW (INCLUDING VARIATION LSFNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LSFNW (unarmored) and LSFNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated. (See table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene or ethylene propylene rubber insulation, colored black. (See table I for thickness.) Standard identification code applied by method 1.
Fourth - Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used to form a firm, well-rounded assembly.
Fifth - Binder tape applied helically with overlap.

Construction variant for type LSFNW

Sixth - Cross-linked polyolefin jacket. (See table I for thickness). Cable surface marking required.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LSFNWA

Sixth - Cross-linked polyolefin jacket. (See table I for thickness).
Seventh - Braided metal armor and paint.

TABLE I. Construction details.

Military 1/ part no. M24643/50	Size 2/	AWG conductor size	Insulation thickness nominal (inch)	Cable jacket thickness nominal (inch)	Overall diameter (inches)			
					Type LSFNW		Type LSFNWA	
					min	max	min	max
-01UN	3	16(class B)	0.030	0.040	0.413	0.447	0.463	0.497
-02UN	4	14(class B)	.030	.040	.475	.513	.525	.563
-03UN	9	10(class B)	.030	.040	.583	.630	.635	.680
-04UN	23	7(class B)	.045	.050	.768	.830	.818	.880

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSFNW and LSFNWA.

TABLE II. Electrical details.

Size 1/	AWG conductor size	Conductor resistance per 1000 feet at 25°C (max) (ohms)	Insulation resistance/ 1000 feet (minimum) (megohms)	Voltage withstand (minimum) (volts)	Cold bending mandrel diameter (maximum) (inches)	Accelerated service loading current (amperes)
3	16(class B)	4.3	6000	4000	4	---
4	14(class B)	2.55	5000	4000	4	---
9	10(class B)	1.08	3800	4000	5	---
23	7(class B)	0.543	3700	5500	8	115

1/ Type LSFNW and LSFNWA.

EXAMINATION AND TESTS:

Requirements

Basic electrical:
Conductor resistance - ohms/1000 feet, at 25°C, maximum. (see table II)
Voltage withstand - volts, root mean square, minimum.... (see table II)
Insulation resistance - megohms/1000 feet, minimum..... (see table II)
Conductor continuity..... No failure

Group A:
Visual and dimensional..... No failure

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Cold bending, cable - (See table II for mandrel diameter)	No damage
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Cross-linked polyethylene	
Tensile strength - lbf/in^2, minimum	1800
Elongation - percent, minimum	250
Ethylene propylene rubber	
Tensile strength - lb/in, minimum	700
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lbf/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Cross-linked polyethylene	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Ethylene propylene rubber	
Air oven	
Tensile strength - percent of unaged, minimum	75
Elongation - percent of unaged, minimum	75
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (conductor) - cycles, minimum	25
Permanence of printing (jacket) type LSFNW only - cycles, minimum	250
Cable filler removability	No failure
Heat distortion - percent of unaged, maximum	30
Armor - conformance to materials, construction and coverage, type LSFNWA only	No failure

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QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

	Requirements
Cold working (cable) (minus 20 ± 2°C)	No damage
Accelerated service (type LSFNW-23)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSFNW-3	2500
LSFNW-4, 9	2000
LSFNW-23	1500

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-50)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

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